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A Review of Sensor-Array Magnetic Field Imaging for Metal Defect Detection Using Amplitude, Phase and Machine Learning

Marneedi Suprithi
CB.EN.U4ECE23228
Department of Electronics and Communication Engineering, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Coimbatore, India

***Abstract*—Magnetic sensor arrays are increasingly used to image defects in metals, but leakage-field and current-injection excitation, phase measurement and machine learning are rarely combined. We sorted 44 journal articles from 2021–2026 into a five-branch framework and compared them by sensor, excitation, phase use and interpretation. Of 36 inspection studies, 23 use arrays, six use phase (one with an array) and none reports both excitations. Machine learning mostly classifies or sizes defects. We identify five open challenges.**

***Index Terms*—defect detection, giant magnetoresistance, machine learning, magnetic field imaging, nondestructive testing, phase measurement, sensor array**

I. Introduction

CRACKS, corrosion pits, scratches and holes in a metal part disturb the magnetic field around it, and nondestructive testing (NDT) uses that disturbance to find them. A magnetic field imaging system does this with an array of sensors: it samples the field over the surface and the map is read as an image of the defect. Two kinds of excitation are common. In the leakage magnetic field (LMF) method the specimen is magnetised and flux leaks out around a defect [1]. In current injection (CI) a current is driven through the specimen, and in eddy current testing (ECT) a coil induces it [2]. In both, the defect bends the current path, and we group the two as CI.

Three lines of work have developed side by side. Array probes now have sub-millimetre pitch, from tunnelling magnetoresistance (TMR) arrays at 0.5 mm [3], [4] to unpackaged Hall arrays at 255 µm [5]. Phase measurements improve sensitivity and are less affected by lift-off [6], [7]. Machine learning (ML) has moved from classifying signals to sizing defects [8], [9]. These lines are seldom joined. The reviews we found are organised by technique, such as deep learning for pipeline leakage inspection [8] and pulsed eddy current testing [10], or by defect scale [11]. None of them puts array imaging, excitation method, phase and ML side by side, and that is the comparison someone designing a new array system needs. This missing comparison is the gap we address.

This review makes three contributions:

- a five-branch framework that places each of 41 primary studies by sensing array, excitation, signal quantity and interpretation (Section III);
- a comparison of all 41 studies on the same parameters (Table IV), a branch-level comparison of advantages and shortcomings (Table II) and a comparison of sensing elements (Table III), with counts over the 36 inspection studies;
- five open challenges, each tied to an empty cell in those comparisons (Section VI).

The rest of the paper is organised as follows. Section II describes how we searched and screened. Section III presents the framework, Section IV reviews each branch and Section V gives the comparison results. Section VI discusses the open challenges and Section VII concludes.

TABLE I
Review Protocol and Screening Counts

Criterion	Specification
Window	2021–2026, journal articles only
Included	magnetic-sensor imaging or arrays for defects in metals; phase or quadrature use; ML on magnetic signals; source-localisation methods with arrays
Excluded	proceedings, non-magnetic sensing, batteries and biomedical imaging, records without a readable abstract or a verifiable publisher
Counts	222 identified; 171 excluded on title and abstract; 7 excluded at eligibility; 44 included

II. Research Methodology

A. Research Questions

We asked three questions. RQ1: which array sensors and excitation methods have been used to image defects in metals, and how do LMF and CI compare? RQ2: is phase information used, and with arrays? RQ3: does ML go beyond classifying or sizing defects?

B. Sources, Search Strings and Criteria

We searched OpenAlex in October 2026. It returned records from IEEE, Elsevier, MDPI, IOP and Taylor & Francis journals, among others. We ran and logged twenty title-and-abstract queries. Three did most of the work: *"magnetic flux leakage" AND (array OR imaging) AND (sensor)*, *"eddy current" AND "array" AND (crack OR defect) AND imaging* and *"eddy current testing" AND phase AND (crack OR defect)*. The rest combined the sensor types (TMR, GMR, Hall), "magnetic field imaging" and ML terms. We added the 19 PDFs supplied by the author as seeds. For every record we resolved the DOI on Crossref and copied the bibliographic fields from the returned entry. We kept journal articles because the course rubric asks for them, and set the window to 2021–2026 to match its five-year rule. Table I gives the inclusion and exclusion rules and the screening counts.

C. Reading Depth and Limitations

The 44 included works are 41 primary studies and 3 reviews. We read abstracts for all of them, and also the first pages of the 11 that were among the supplied PDFs. We did not read all 44 in full, so what we say about a paper is limited to what its abstract reports. When we write that no study reports a feature, we mean that no abstract does. One person did the screening and only one index was searched, so some relevant journals may be missing.

III. Proposed Framework

We grouped the works by what they add to a magnetic imaging chain, not by year (Fig. 1). Branch A holds the array probes. Branches B and C split the two excitation families, because an LMF image and a CI image answer different questions: leakage shows where flux is diverted, while an injected current also carries phase. Branch D holds the ML that interprets the signals. Branch E holds array and phase methods from magnetic source localisation and quantum magnetometry. These are not defect studies, but they use inverse-problem tools that the defect papers have not yet picked up. Each work sits in the branch of its main contribution, although a few could fit two. Section V compares the works on the sensing element, the number of elements, the excitation, the use of phase or the quadrature component, and the use of ML.

IV. Core Technological Review

Two relations underlie most of the works below. A straight conductor carrying a current I produces, at a perpendicular distance r, the field

$$B = \frac{\mu_0 I}{2\pi r}, \tag{1}$$

so by (1) a current-carrying wire is a known source for calibrating a sensor. A phase-sensitive receiver, such as the in-phase/quadrature demodulator in [2], returns two components I_c and Q_c of the sensor signal. From these,

$$A = \sqrt{I_c^2 + Q_c^2}, \qquad \varphi = \operatorname{atan2}(Q_c,\, I_c) \tag{2}$$

give the amplitude A and the phase φ relative to the excitation, as in (2).

A. Array Imaging Probes (Branch A)

Among the 36 inspection studies, TMR is the most common sensing element (8 studies), followed by Hall (5), giant magnetoresistance (GMR, 3) and anisotropic magnetoresistance (AMR, 2). Pitch is the main design variable. TMR arrays at 0.5 mm pitch image vector field components [3]. Staggering two rows halves the pixel pitch to 0.25 mm, and with this [4], [23] detect a $1 \times 0.1 \times 0.1$ mm defect. Unpackaged Hall elements reach 255 µm pitch [5], compared with 1.2 mm for a 16×16 Hall array that holds the sensor within 0.5 mm of the surface [12]. Finer pitch costs channels and scan time, so [24] simulates pitches from 100 µm to 5 mm to find how few sensors still keep a flaw detectable.

Packaging follows the part being inspected. A flexible TMR probe bends to a 5.5 mm radius and finds defects buried 3 mm deep [25]. An elliptical probe fits 5 mm tubes [26]. A tower-type transmitting coil raises excitation uniformity to 98.02% for wide-area imaging [27]. Among the abstracts we read, channel consistency is treated explicitly only in [13], where a spatially dependent calibration compensates offset and responsivity differences between eight GMR channels. Single-element probes also exist for curved or fine-scale work. A miniature GMR gradiometer images surface and subsurface flaws by scanning [28], and flexible AMR sensors with an excitation wire detect defects down to 30 µm deep and 400 µm wide at a signal-to-noise ratio of 3 [29]. Sparse arrays can be paired with an inverse method: an 8×8 array with 4 mm spacing resolves 2 mm defects using a variational Bayesian solver [30].

B. Leakage Magnetic Field Methods (Branch B)

LMF studies magnetise the specimen with a permanent magnet [1], [31], [32] or a yoke [14], and no abstract in this branch reports phase. The work goes into raising the signal and removing nuisance variables. A pot-shaped yoke raises the peak-to-peak signal to 1.93 times that of a U-shaped one [14], and a filter built from the x and z components suppresses lift-off on rail heads [15]. Arrays give geometry from a single scan. A circular array gives crack angle, width and length with 95.0%, 91.2% and 88.4% accuracy [16]. Fifteen three-axis Hall sensors feed pseudo-colour images to a particle-swarm-optimised least-squares support-vector machine [33]. Three sensor-head designs, one of them circular for small pipes, are compared in [1], which also sets a threshold at the 99.99 confidence level.

C. Current Injection and Phase Methods (Branch C)

CI and eddy-current methods can use phase, and the evidence for doing so agrees across papers. The phase of the receiving-coil signal is less susceptible to lift-off variation, and a 1 MHz two-coil sensor detects defects under 500 µm in titanium alloys [6]. In a flexible eddy-current sensor, phase follows the excitation frequency, is more sensitive at higher frequency and needs a correction for large cracks [7]. A phase-difference method separates fatigue cracks from thread signals in bolts [34]. When ML is added, phase can go into the network. A complex-valued CNN reconstructs defect shapes from amplitude and phase without baseline data [17], and a portable I/Q device produced 48,000 scans for depth classification [2]. The array studies in this branch mostly stay with amplitude. They use differential array coils with Mask-RCNN [35], hand-held arrays with Hough-transform analysis [36] and pulsed arrays for multiple defects [37]. Only [38] images the quadrature component, with a TMR and coil array.

D. Machine-Learning Interpretation (Branch D)

In these works ML maps a magnetic signal or image to a defect class or size. YOLOv5-based detectors are trained on leakage images, with COMSOL simulation used to enlarge the data [18]. A cascaded detector and quantifier reach maximum errors of 1.30, 1.65 and 0.47 mm in length, width and depth [39], and a simulated-annealing-optimised support-vector regressor sizes tubing defects [40]. A further step is to put physics into the training. Schematically, the training objective in (3) becomes

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda\, \mathcal{L}_{\text{phys}}, \tag{3}$$

where $\mathcal{L}_{\text{data}}$ is the usual data misfit, $\mathcal{L}_{\text{phys}}$ penalises violation of a physical model and λ weights the two. In [9] the physical term encodes MFL quantification theory, and the network quantifies length, width and depth jointly, with depth reported as the main gain. In [19] it is the magnetic dipole model, and the method is validated on synthetic data only. A review of deep learning for pipeline MFL names interpretable models and data augmentation as the main future needs [8]. ML appears in 5 of the 23 array studies, for example AMR imaging features [41] and Hall-network regression [1].

E. Source Localisation and Phase Beyond NDT (Branch E)

These five works are not defect studies, but they bear on the inverse problem. Nine planar Hall sensors with a fully connected network map a magnetic source better than interpolation or Levenberg–Marquardt optimisation, and simulation suggests that more than 50 sensors would be needed without ML [20]. A super-resolution network sharpens array data before inversion and locates three to five magnets with under 3 mm error in under 300 ms [21]. Subtracting the background across array points allows multipoint tracking [42], and satellite sensors let an attention-based network reconstruct and cancel noise [43]. In NV magnetometry, mapping the resonance shift to a phase estimate is faster than Lorentzian

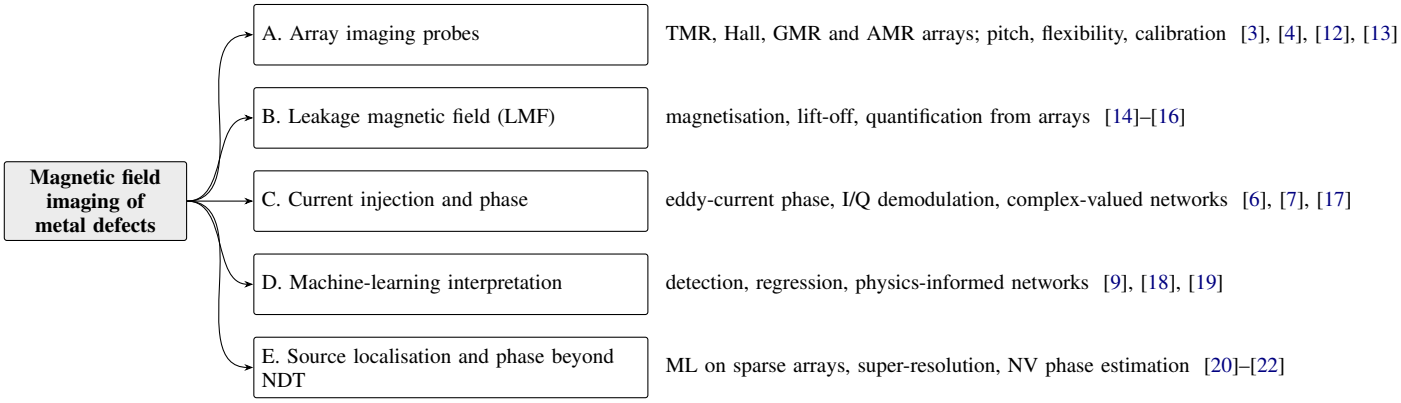


Fig. 1. Framework used to organise the 41 primary studies. Branch sizes: A 15, B 8, C 8, D 5, E 5.

fitting [22]. Together they show that a sparse array with a learned inversion can stand in for dense sensing, the opposite trade-off to the race for finer pitch in Branch A.

V. Comparative Performance and Synthesis

We compare the works at three levels: branch against branch (Table II), sensing element against sensing element (Table III), and study by study (Tables IV and V). Results are quoted as each abstract reports them. They come from different specimens, defect sizes and metrics, so none of the result columns is a ranking. A dash under Ph. or ML in the study tables means the abstract does not report that feature. We counted a study as using phase when its abstract or title reports phase, a phase difference, or the quadrature component of the signal.

A. Comparison Across Branches

The clearest difference between the branches is the material each excitation is used on. LMF studies address steel: rail heads, tank floors, pipelines, plates and galvanised sheet [14], [15], [31], [33], [39]. CI studies also cover aluminium, titanium alloys, stainless steel and nonferrous multilayer plates [2], [3], [6], [30]. The two families are tested on different materials, which is why no study shows which is better for a given defect. They also differ in the signal. LMF work takes geometry from amplitude patterns and leaves interpretation to ML [16], [18]. CI work has the extra phase channel but uses it mostly in single probes [6], [7]. Finally, the role of the array differs. Branch A and B studies buy resolution with more elements, while Branch E shows that a learned inversion can replace elements [20].

B. Comparison of Sensing Elements

Table III compares the four element types in the inspection studies. TMR is the most common. At low excitation frequency it is more sensitive than an induction coil [44], and the TMR arrays in the study tables reach 0.5 mm pitch. Unpackaged Hall elements give the finest reported pitch [5]. GMR and AMR appear in fewer arrays, and the one explicit treatment of channel mismatch is for GMR [13]. Element type only partly predicts excitation. TMR is used with CI in seven of its eight studies and with LMF in one [31], while all five Hall studies use LMF.

C. Study-by-Study Comparison

Tables IV and V list all 41 primary studies on the parameters of the framework, with the specimen or scope of each evaluation in the last column. The 41 studies are spread over 2021–2025 at 6 to 11 per year (Fig. 2).

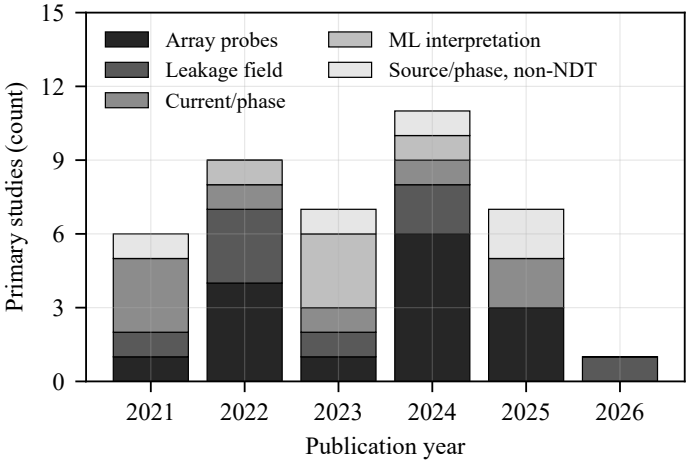


Fig. 2. Primary studies per year by framework branch (A–E as in Fig. 1); 2026 is a partial year.

D. Findings and Answers to the Research Questions

Counting over the 36 inspection studies (Branches A–D) gives four findings.

Arrays are the norm, phase is not. Twenty-three of the 36 use an array, but only six use phase or the quadrature component, and of those only [38] uses an array. The other phase evidence [2], [6], [7], [17], [34] comes from single probes.

LMF and CI are studied apart. Fifteen studies use LMF and twenty use CI. No abstract reports both on the same specimens, so the choice between them rests on convention, not measurement.

ML interprets signals, not fields. Twelve of the 36 use ML, five of them with arrays and two with phase [2], [17]. Their outputs are classes or sizes. One work [19] is validated on synthetic data only.

Resolution claims are not comparable. The smallest reported defects range from $1 \times 0.1 \times 0.1$ mm [4] to 400 µm [31] and 2 mm [30]. They were measured on different materials, lift-offs and criteria, and no specimen is shared between studies.

For RQ1, TMR, Hall, GMR and AMR arrays are all used, TMR most often, with CI on nonferrous and LMF on ferromagnetic specimens and no direct comparison between them. For RQ2, phase is used in six inspection studies and with an array in one. For RQ3, ML mostly classifies or sizes defects, and none of the reviewed works estimates the underlying current pattern.

VI. Discussion: Open Challenges and Future Research Directions

The findings of Section V leave five problems open. Each comes from a count or an empty cell in the comparison tables.

TABLE II
BRANCH-LEVEL COMPARISON OF APPROACHES

Branch	Approach	Main advantages	Shortcomings	Application
A. Array probes	TMR, Hall, GMR and AMR arrays, fixed or scanned	one scan gives an image; pitch down to 255 µm [5]; $1 \times 0.1 \times 0.1$ mm defects detected [4]; flexible forms [25]	channel consistency needs calibration [13]; Hall array signal depends on keeping the distance below 0.5 mm [12]; phase used in one array study [38]	small, buried and curved defects; tubes
B. Leakage field	static magnetisation by magnet or yoke, array reads leakage	simple excitation by a permanent magnet and speeds up to 1200 mm/s [31]; crack angle, width and length from array geometry [16]	lift-off interference [15]; no phase reported; steel specimens in most studies	rail, pipes, tank floors, plates
C. Current injection and phase	injected or induced current; amplitude and phase	phase less sensitive to lift-off [6]; nonferrous metals covered [3], [30]; deep learning on I/Q data [2]	5 of 8 studies use a single probe; phase depends on excitation frequency [7]	cracks, bolts, welds, rivet joints
D. ML interpretation	CNN, YOLO-type detectors, regressors, physics-informed networks	joint length, width and depth estimation [9]; 98.9% detection precision [39]	synthetic data only in one study [19]; simulated augmentation [18]; outputs are classes or sizes	pipelines, tubing
E. Source localisation	learned inversion on sparse arrays; phase estimation	more than 50 sensors needed without ML [20]; error under 3 mm in under 300 ms [21]	not defect studies; the sensor-count result is from simulation	magnets, capsule tracking, NV magnetometry

TABLE III
COMPARISON OF SENSING ELEMENTS IN THE 36 INSPECTION STUDIES

Element	n	Reported configurations	Reported strength or limit
TMR	8	64 elements [44]; 0.5 mm pitch, 0.25 mm pixel pitch [4]; 16 flexible elements [25]	more sensitive than coils at low frequency [44]; bare dies wire-bonded to a PCB [4]
Hall	5	256 elements at 255 µm [5]; 16×16 at 1.2 mm [12]; 15 three-axis [33]	distance kept below 0.5 mm [12]
GMR	3	8 channels [13]; flexible in-line array [32]; gradiometer [28]	offset and responsivity mismatch needs calibration [13]
AMR	2	array with ML [41]; flexible with wire [29]	30 µm deep, 400 µm wide at SNR 3 [29]

1. LMF against CI on the same defects. Fifteen studies use LMF and twenty use CI, and no abstract compares them on the same specimens. One array system that can switch between magnetisation and current injection, scanned over the same cracks, scratches, holes and corrosion, would show which excitation suits which defect.

2. Phase with arrays. Phase helps in single probes [6], [7], but [38] is the only array study that uses the quadrature component. Reading phase from many multiplexed channels is the open problem. It also needs the channel calibration that [13] found necessary even for amplitude alone.

3. From classification to current reconstruction. In the reviewed works ML returns classes, sizes or positions. None estimates the current pattern behind a field image. Source-localisation work shows that a learned inversion can stand in for dense sensing [20], [21], and physics-informed training [9], [19] is a natural way to apply it here.

4. A common benchmark. Smallest-detectable-defect figures run from 400 µm to 2 mm and cannot be compared. A shared set of specimens, a fixed lift-off and a stated detection criterion would let array designs be ranked.

5. Data for learning. One network is validated on synthetic data alone [19] and another relies on simulated augmentation [18]. The review in [8] lists augmentation and interpretability as needs. Open measured datasets like the one released with [2] are rare for arrays.

VII. CONCLUSION

We organised 44 journal articles from 2021–2026 into a five-branch framework for magnetic field imaging of metal defects and compared them on the same parameters. Array probes with sub-millimetre pitch, phase measurement and learned interpretation have each moved forward, but they are rarely used together. Of the 36 inspection studies, 23 use arrays, six use phase and only one does both, and none compares LMF with CI. ML mostly classifies or sizes defects and does not recover the current pattern. Resolution claims cannot be compared because no common specimen exists. We expect work to move toward switchable excitation, phase-sensitive arrays and physics-informed inversion, tested on shared specimens and open datasets.

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TABLE IV
COMPARISON OF THE PRIMARY STUDIES, BRANCHES A AND B (BR.: FRAMEWORK BRANCH; EXC.: EXCITATION; PH.: PHASE OR QUADRATURE COMPONENT USED; ML: MACHINE LEARNING USED; NS: NOT STATED)

Work	Br.	Sensor (elements)	Exc.	Ph.	ML	Main result as reported	Application and scope
[44]	A	TMR, 64 elements	CI	–	–	$6 \times 3 \times 1$ mm defect under 10 mm insulation; $9 \times 3 \times 1$ mm under 20 mm	defects under thick insulation; machined rectangular defect
[3]	A	TMR, 3 arrays × 32	CI	–	✓	x/y/z images; k-means classes defects, linear regression gives orientation	aluminium samples with machined defects
[4]	A	TMR, 2 staggered rows, 0.5 mm pitch	CI	–	–	$1 \times 0.1 \times 0.1$ mm defect; pixel pitch 0.25 mm	small-defect inspection with a staggered two-row probe
[23]	A	TMR, 2 staggered rows	CI	–	–	$1 \times 0.1 \times 0.1$ mm defect on stainless steel; pixel pitch 0.25 mm	small defect on a stainless-steel sample
[25]	A	TMR, 16 elements, flexible PCB	CI	–	–	defects 3 mm below surface; bend radius 5.5 mm	buried defects in curved, cylindrical samples
[26]	A	TMR, 16 elements, elliptical PCB	CI	–	–	tubes of 5 mm inner diameter	small-diameter tubes of nuclear flux tubes
[12]	A	Hall, 256 elements (16×16)	LMF	–	–	19.2×19.2 mm area; 1.2 mm separation; distance below 0.5 mm	magnet, phone coil array, magnetised steel plate; no defect
[5]	A	Hall, 256 elements linear	LMF	–	–	255 µm pitch; 256×256 pixel images; 99.61 ppi	magnetised scissors, phone charging coil; defects only as potential
[41]	A	AMR array	NS	–	✓	six metal samples; ML-generated imaging features	six metal samples; excitation not stated
[13]	A	GMR, 8 channels	CI	–	–	spatially dependent calibration; defect spacing above 2 mm	metal sheet; channel calibration needed
[28]	A	GMR gradiometer, single probe	CI	–	–	surface and subsurface flaws imaged by scanning	metallic samples with machined defects; single probe
[29]	A	flexible AMR with excitation wire	CI	–	–	depth to 30 µm, width to 400 µm at SNR 3	curved and flat aluminium samples; benchmarked against coils and rigid GMR
[30]	A	magnetic sensors, 8×8, 4 mm pitch	CI	–	–	2 mm defects; 0.5 mm depth resolution; variational Bayes inverse	multilayer nonferrous plates; simulation plus prototype
[27]	A	magnetic sensing array, tower coil	CI	–	–	uniformity 98.02%, 43.82% above single loop	wide-area imaging of metal workpieces
[38]	A	TMR array plus array coils	CI	✓	–	quadrature-component image; decision-level fusion	steam generator tube mock-up
[1]	B	Hall network, 3 head designs	LMF	–	✓	GPR, MDSS, KNN; threshold at 99.99 confidence	flat components and small pipes; metal loss
[24]	B	QWHE Hall, simulated pitch	LMF	–	–	pitch 100 µm to 5 mm simulated; 400 Hz, 30 mT RMS	ground mild steel weld sample, two flaws; pitch simulated from one scan
[16]	B	circular array	LMF	–	–	accuracy 95.0% angle, 91.2% width, 88.4% length	rail top surface cracks
[33]	B	3-D Hall, 15 sensors	LMF	–	✓	pseudo-colour imaging; PSO-LSSVM	tank floor steel plate
[32]	B	flexible GMR array	LMF	–	–	in-line robot; ANSYS Maxwell lift-off optimisation	steel pipes; in-line robot with permanent magnet
[31]	B	TMR, single, 300 mV/V/Oe	LMF	–	–	400 µm zinc dross; 0 to 1200 mm/s	hot-dip galvanised sheet; zinc dross
[15]	B	sensor array	LMF	–	–	lift-off suppression from x and z difference	rail head surface defects
[14]	B	sensor array	LMF	–	–	peak-to-peak up to 1.93 times U-shaped yoke	steel plate

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TABLE V
COMPARISON OF THE PRIMARY STUDIES, BRANCHES C, D AND E (NA: NOT APPLICABLE; OTHER ABBREVIATIONS AS IN TABLE IV)

Work	Br.	Sensor (elements)	Exc.	Ph.	ML	Main result as reported	Application and scope
[2]	C	ECT probe, single	CI	✓	✓	48,000 scans; 18 defects; ResNeXt 93.58%, 0.3 to 2.0 mm at 0.1 mm	single probe; stainless steel sheet, depth classes
[7]	C	flexible eddy current sensor	CI	✓	–	phase mainly follows excitation frequency; phase correction for large cracks	aluminium plates with artificial cracks; single sensor
[6]	C	two-coil, 1 MHz	CI	✓	–	phase less sensitive to lift-off; defects under 500 µm	titanium, titanium aluminide and stainless steel; single coil pair
[34]	C	differential probe	CI	✓	–	phase difference; thread-lift-off equivalence	titanium alloy aircraft bolts; thread-signal suppression
[35]	C	flexible differential array coil	CI	–	✓	four modes at 1 MHz; Mask-RCNN	weld inspection; flexible probe on uneven weld surface
[36]	C	handheld eddy current array	CI	–	–	Hough circle transform and texture analysis	aircraft rivet joints; simulated sample
[17]	C	impedance signals	CI	✓	✓	complex-valued CNN uses amplitude and phase	carbon steel; defect reconstruction
[37]	C	pulsed eddy current array	CI	–	–	widths within one scan step	large-area metal components; multiple defects
[18]	D	MFL images	LMF	–	✓	YOLOv5-based; COMSOL augmentation	MFL small-defect detection; simulated augmentation
[9]	D	MFL, three-axis data	LMF	–	✓	physics-informed; length, width, depth together	MFL defect size estimation
[19]	D	synthetic MFL data	LMF	–	✓	dipole-model constraint; synthetic data only	in-pipe defect sizing; synthetic data only
[39]	D	three-axis MFL sensor on robot	LMF	–	✓	98.9% precision; errors 1.30, 1.65, 0.47 mm	steel pipelines; in-line inspection robot
[40]	D	MFL, tubing	LMF	–	✓	SA-MLSSVR regression	oil and gas tubing; wellhead testing
[20]	E	planar Hall, 9 sensors (3x3)	NA	–	✓	FCNN beats interpolation and LMA; over 50 sensors needed without ML (simulation)	magnetic source location; sensor-count claim from simulation
[21]	E	magnetic array	NA	–	✓	3 to 5 targets; error under 3 mm; under 300 ms	multi-magnet localisation; array field data
[42]	E	magnetic sensor array	NA	–	–	background subtracted; multipoint localisation by optimisation	wireless capsule endoscope tracking; not a defect study
[22]	E	NV widefield	NA	✓	–	phase estimation from ODMR; faster than Lorentzian fitting	widefield NV magnetometry; not a defect study
[43]	E	auxiliary sensor array	NA	–	✓	attention-LSTM noise reconstruction	magnetic signal denoising; not a defect study

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